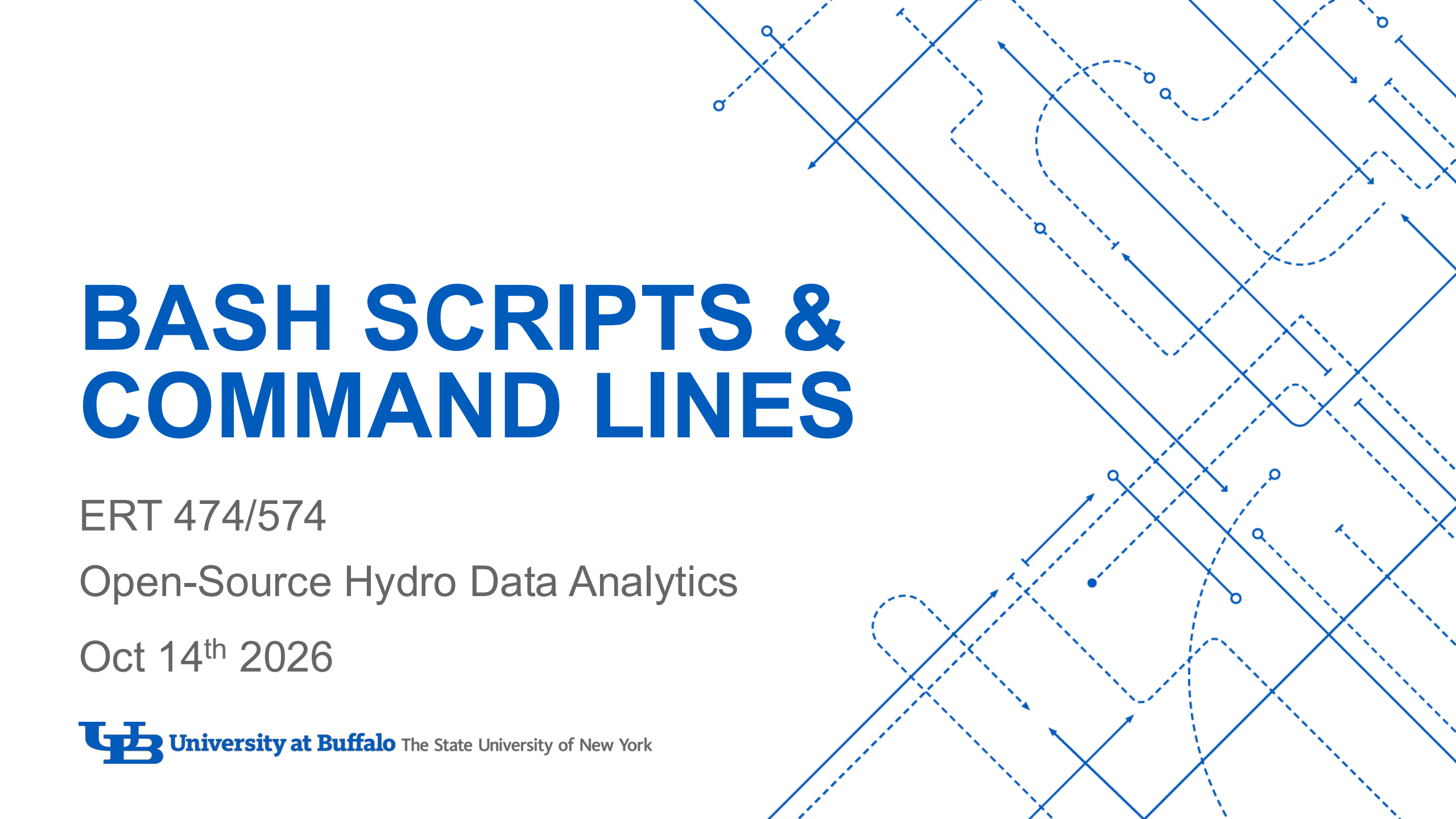


BASH SCRIPTS & COMMAND LINES

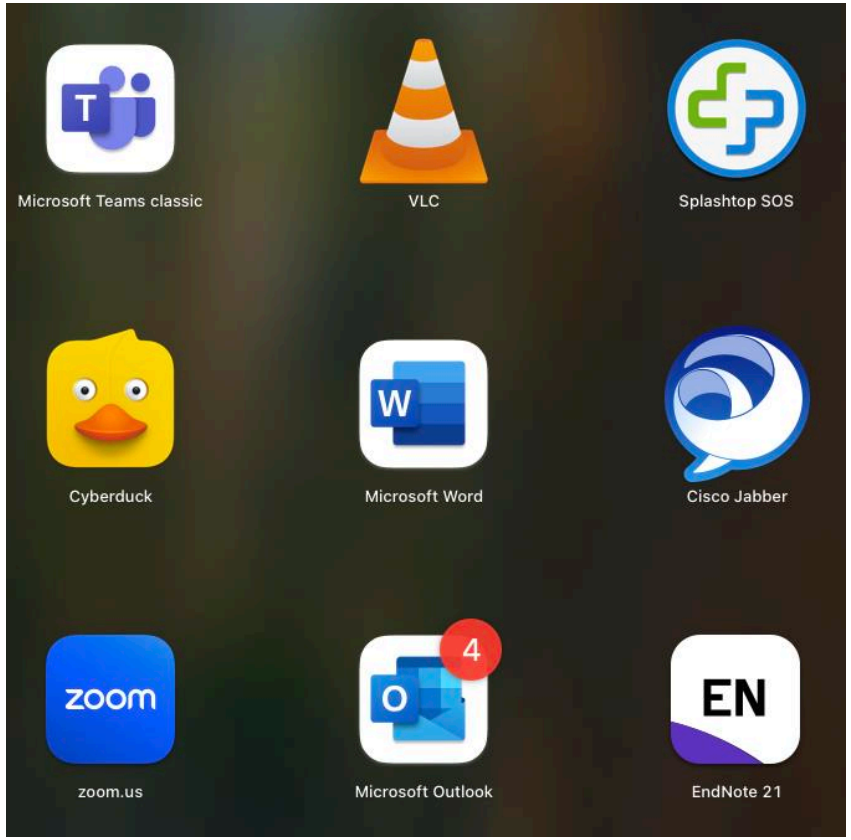
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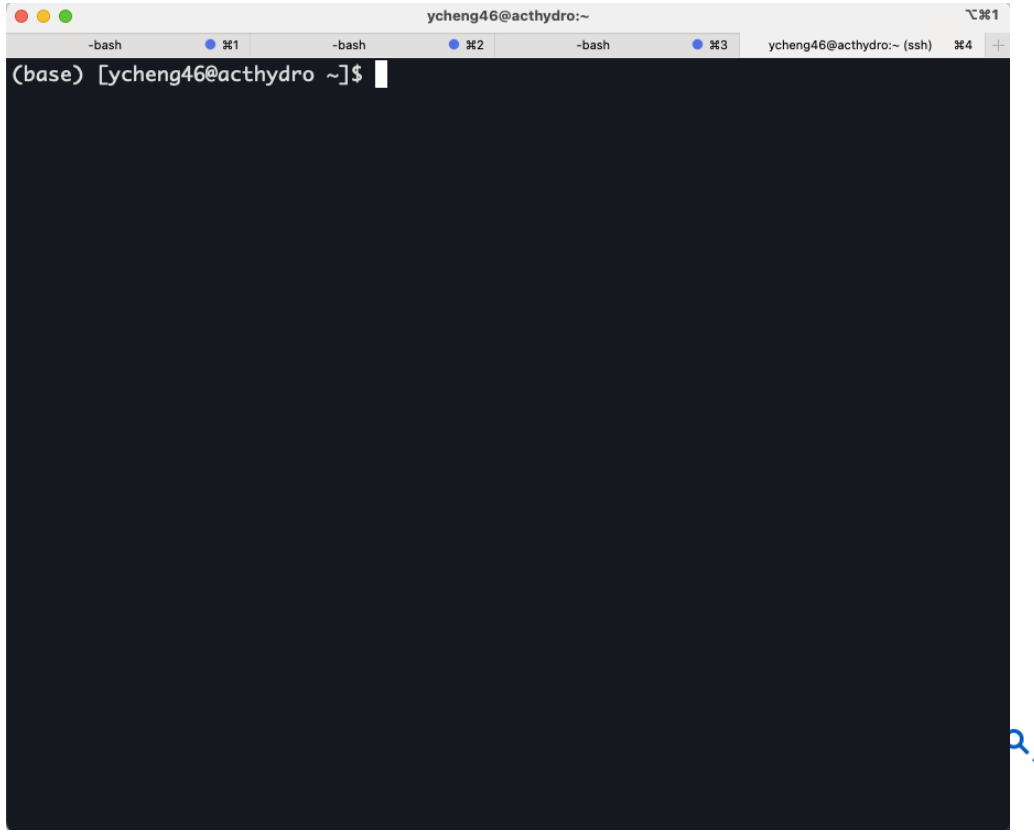
Open-Source Hydro Data Analytics

Oct 14th 2026

University at Buffalo The State University of New York

GUI versus Command Lines

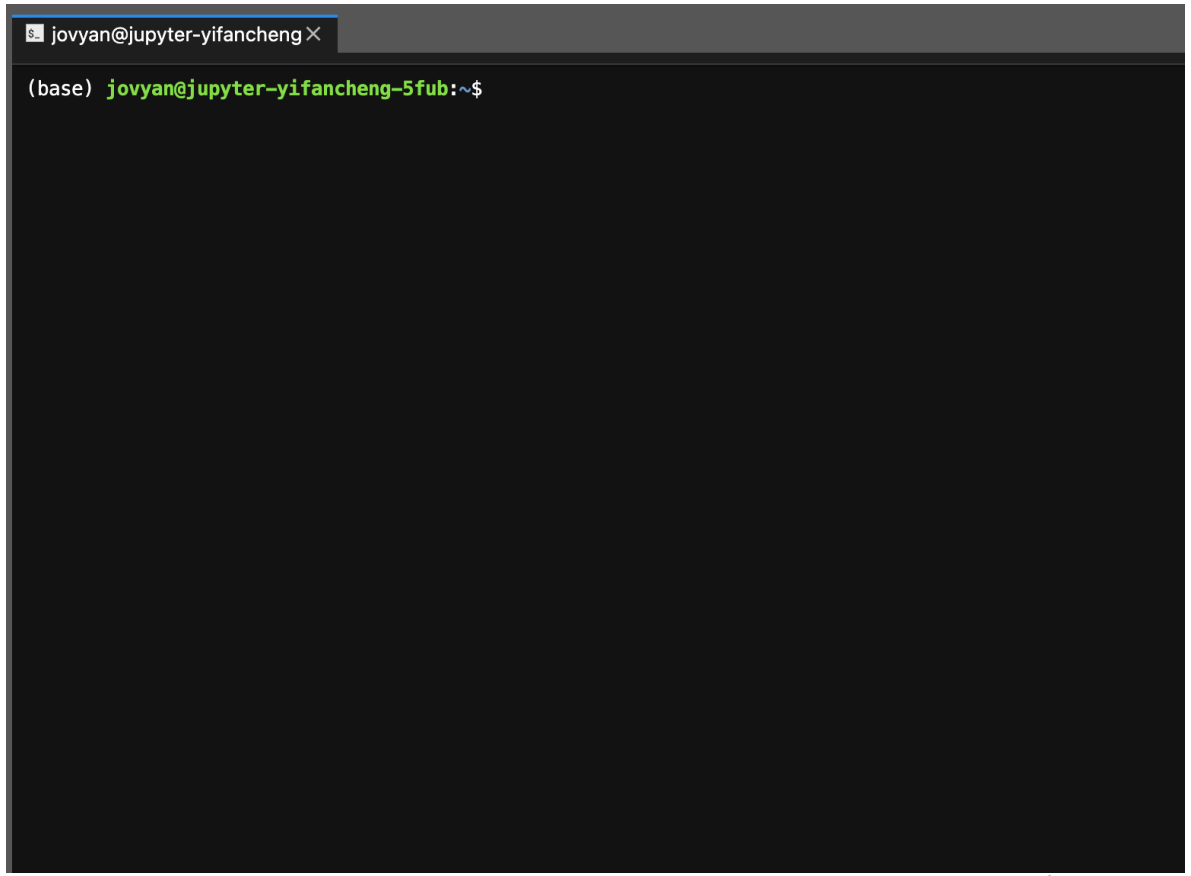




What is the differences between the Graphical User Interface (GUI) and the Command Line?

What is bash script?

- A bash script is a file containing a sequence of commands that are executed by the bash program line by line. It allows you to perform a series of actions, such as navigating to a specific directory, creating a folder, and launching a process using the command line.



Why is it important to learn Bash?

Automation

Allow you to automate repetitive tasks and processes, saving time and reducing the risk of errors that can occur with manual execution.

Portability

Can be run on platforms and operating systems

- Unix
- Linux
- macOS
- Windows (emulators or virtual machines)

Flexibility

Highly customizable and can be easily modified to suit specific requirements.

Accessibility

Easy to write and don't require any special tools or software. They can be edited using any text editor, and most operating systems have a built-in shell interpreter.

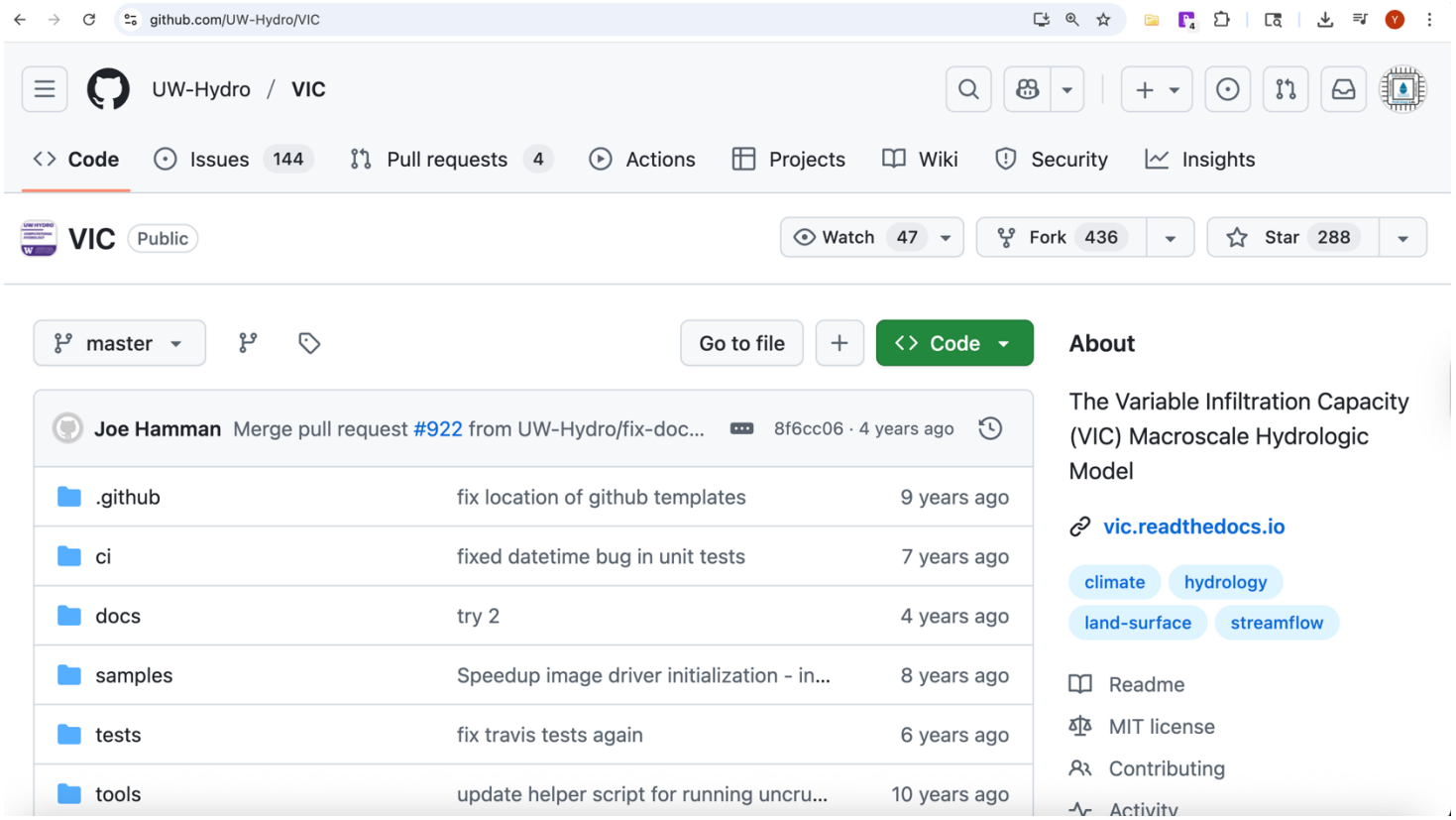
Integration

Can be integrated with other tools & applications

- databases
- web servers
- cloud services

allowing for more complex automation & system management tasks.

Open-Source Hydrologic Model



In this class, we will learn how to configure and run the Variable Infiltration Capacity (VIC) hydrology model!

Here, you were provided the source code. What is your next step?

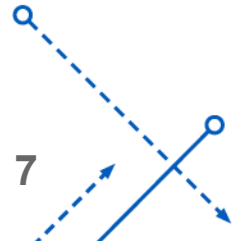
We need to use Command Lines to run VIC



Before we can run the models in GitHub Codespaces, we need to learn how to navigate the system.

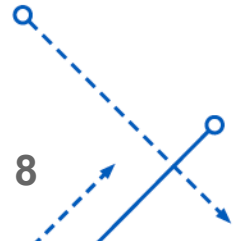
Commonly Used LINUX Utilities

- `cd` shell command for changing directory
- `head` displays the first n lines of data; if no “n” specified, it shows 10 lines
- `tail` display last n lines of file, if no “n”, the last 10 displayed
- `ls` lists files in the current directory
- `mkdir` creates a directory
- `more` allows you to scroll through a list of files one page at a time
- `mv` rename file; overwrite file; move to new location
- `pwd` print current working directory
- `rm` delete file
- `rmdir` delete directory
- `wc` counts lines, words, and characters



Text Searching

- `grep` useful utility for searching a file
- finds lines with the letter **q** in a file
  - `grep q filename`
- finds lines with the letter **q** and gives the line number in the file
  - `grep -n q filename`



LINUX Utilities

- `cat` takes input from standard in or list of file(s) and displays them to standard output
- To use, type
  - `cat filename`
- Then press the return key.